Evolutionary Algorithms for Improving Chain-of-Thought Reasoning in Vision-Language Models

Project Report

APMTH 207: Advanced Scientific Computing: Stochastic Methods for Data Analysis, Inference and Optimization

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December 2, 2024

Abstract

Vision-Language Models (VLMs) have demonstrated remarkable advancements in understanding and reasoning across multimodal tasks but often struggle with complex, visually ambiguous scenarios, leading to inconsistencies and errors. While Chain-of-Thought (CoT) prompting has shown potential in improving reasoning for large language models, its application to VLMs remains underexplored. Furthermore, existing CoT methods typically rely on static prompts, which can degrade performance in tasks requiring nuanced, context-specific reasoning. This study introduces an Evolutionary Chain-of-Thought (EoT) prompting framework that dynamically optimizes CoT prompts through evolutionary algorithms tailored to individual tasks. Using GPT-4o, a state-of-the-art multimodal model, we evaluate our approach on the challenging Japanese Uncertain Scenes (JUS) dataset, which features visually ambiguous images and complex prompts. The EoT framework enhances reasoning accuracy by iteratively refining prompts through our proposed crossover and dual-mutation operations guided by performance metrics, including semantic similarity and numeric accuracy. Results indicate that EoT prompting significantly outperforms both baseline and static CoT methods, achieving improved accuracy and reduced variance in performance across tasks such as object counting and image recognition. By addressing reasoning inconsistencies and leveraging adaptive CoT prompting, this research demonstrates a robust framework for improving VLM reliability in complex scenarios, paving the way for broader industry applications.

1 Introduction

Vision-Language Models (VLMs) have shown impressive progress in interpreting visual content and reasoning across diverse tasks but continue to face challenges when accurately addressing complex problems of object recognition in visually ambiguous scenarios. Though hallucinations remain a problem for all language model types, complex vision-language tasks expose greater inconsistencies in VLM reasoning, leading to incorrect outputs and particularly poor performance.

While Chain-of-Thought (CoT) prompting has shown significant promise in improving reasoning for general large language models by generating the intermediate steps, far less research has been done on CoT applications to vision language models. Additionally, existing CoT approaches often rely on static, identical prompts that may introduce disruptions and degrade overall performance due to the ongoing evolution of sentence prefixes during the pre-training phase (Jin et. al., 2024). We hypothesize that this rigidity will be particularly detrimental in the case of visually complex scenarios that require nuanced element descriptions and context-specific reasoning.

This project focuses on improving VLM performance in challenging vision-language tasks, such as object counting and image recognition, by leveraging CoT reasoning enhanced with evolutionary algorithms to optimize prompts. The main contributions of this paper are: (1) demonstrating how adaptive CoT prompting can improve performance on complex vision tasks, especially in ambiguous settings like those represented in the Japanese Uncertain Scenes (JUS) dataset; (2) showcasing the effectiveness of evolutionary algorithms for prompt optimization, using cross-entropy as the fitness function to ensure linguistic coherence. By addressing these challenges, this research builds upon prior work to deliver a more robust framework for enhancing VLM accuracy in complex scenarios.

2 Background and Notation

To perform the experiment, we select GPT-4o, OpenAI's multimodal model trained end-to-end to integrate text, voice and vision. It has a 128,000-token context window with a knowledge cutoff date in October 2023, and is designed to be quicker and more computationally efficient than previous models. We select GPT-4o as it outperforms GPT-4 on major benchmarks, including simple math, language comprehension and vision understanding, in addition to supporting stronger contextual understanding of idioms and cultural references. Due to resource constraints, we utilize GPT-4o Mini for text-only generation, which is more straightforward and cost efficient.

2.1 Zero-shot Chain of Thought Prompting

Chain-of-thought (CoT) (Wei et al., 2022) is a prompting technique that enhances the reasoning ability of language models by decomposing the original problem into step-by-step, sequential components. Instead of directly producing a final answer, the model is guided to articulate intermediate steps, clarifying the logical or causal connections in the reasoning process and thus mirroring human problem-solving capabilities.

The zero-shot chain-of thought prompting method proposes attaching the same sentence before or after each given prompt. We take inspiration from several prior approaches such as the standard Kojima et al. (2022) which appends "Let's think step by step" to

the target problem, Plan-and-Solve prompting Wang et al. (2023a) and RE2 prompting, Xu et al. (2023) which adds "Read the question again" combined with "Let's think step by step" to the target prompt. In this study, we modify the original prompt Q given by the JUS dataset by appending our zero-shot CoT instruction T. Given an original prompt:

> "How many food items are shown in this photo?"

We obtain the resulting CoT prompt:

> "Please provide a detailed step-by-step reasoning for your answer. Ensure that your reasoning is clear and follows a logical progression."

We include the first line to explicitly instruct the model to break down reasoning into detailed steps to minimize errors. The second line improves interpretability and handling of uncertainty by providing clear explanations. We can thus formalize the zero-shot CoT prompting method as:

$$P(A \mid T, Q) = P(A \mid T, Q, C)P(C \mid T, Q) \tag{1}$$

where Q denotes the prompt or question, T is attached CoT instruction, C represents a sampled rationale in natural language and A is the final generated answer (Jin et. al., 2024).

2.2 Evolutionary Chain of Thought Prompting

While zero-shot CoT has shown significant improvements for language model performance, applying a uniform prompt across every task can compromise predictive accuracy and reduce overall performance, especially in complex VLM scenarios where slight variations in phrasing may result in notable impacts. To remediate this limitation, we turn to evolutionary algorithms (EA), a class of optimization algorithms inspired by natural evolution, to determine the most effective CoT prompt tailored to each image and prompt instance.

EAs simulate evolutionary processes such as selection, mutation, crossover, and survival of the fittest to iteratively improve a population of candidate solutions to a given problem. As proposed by Jin et. al. 2024, the evolutionary chain-of-thought (EoT) prompting method utilizes a LLM as an evolutionary optimizer to perform crossover and mutation operations on two parent CoT promptings to generate a diverse population of prompts. Then, the LLM selects the "most suitable" CoT prompting to integrate into a rewriting of the original given problem, before finally predicting the answer.

We expand upon the EoT framework by implementing advanced crossover and mutation operators specialized for improving high-uncertainty counting and recognition reasoning, using GPT-4o Mini as the evolutionary optimizer. We replace the "black box" LLM-based prompt selection with a defined fitness function, and LLM-Based Crossover with a formulaic method. Moreover, we introduce a novel dual-mutation operation that analyzes previous prompt performance evaluated on ground-truth values to intelligently guide the evolution of CoT prompts.

3 Methodology

3.1 Japanese Uncertain Scenes Dataset

The publicly available (https:// github.com/ML-RUG/jus-dataset) Japanese Uncertain Scenes Dataset (JUS) was designed specifically to be visually challenging with difficult prompts, originally created for a study testing VLMs on image recognition tasks to assess their uncertainty estimation capability. The dataset consists of 39 scenes from Japan (Osaka, Tokyo, Kyoto, Hiroshima) with corresponding image recognition prompts ranging from counting questions to identifying food items, restaurants, cultural sites, urban and natural landscapes, religious locations, and transportation. We perform our study on this dataset due to its difficult-to-answer questions and visually ambiguous images, thus requiring the VLM to follow in-depth reasoning paths to avoid confusion and hallucinations. In addition, the dataset is pre-labeled, matching each image to a specific prompt (question), text answer (ground truth), numeric answer (null if not a counting problem), and question type (counting problem equals True/False).

3.2 Implementation

We assess VLM performance on the JUS dataset using three methods: a baseline approach, zero-shot CoT prompting, and our novel advanced evolutionary CoT method that optimizes prompts through multiple generations.

For the baseline approach, we implement basic image prompting using direct LLM queries without chain-of-thought techniques. The prompt given by the JUS dataset is directly given to GPT-4o and the response is evaluated. For standard CoT, we append the static two-line prompt in Section 2.1 to the given JUS prompts.

3.3 Advanced Evolutionary Prompt Optimization

We initialize two base CoT prompts for each type of problem scenario: 'counting' or 'descriptive'. Each prompt includes an instruction line, i.e. "analyze this image step by step," followed by between 1-5 detailed steps. We also require the LLM to state its confidence level and any uncertainties in addition to addressing the given problem (See detailed base prompts in Appendix). To further extend population diversity, we also append the original JUS prompt, and two additional variations by appending "Please be specific and detailed in your response" and "Analyze the image carefully before answering."

3.3.1 Elite Preservation

In each generation, we keep the highest-scoring 20% of prompts as "elite prompts" which are then used to select parents for the next generation. For a given prompt Q, we randomly select parent prompts T_1 and T_2 from the elite members and apply a crossover operation followed by a LLM-based mutation on the crossovered prompt:

$$T_c = Crossover(T_1, T_2)$$
$$T_m = Mutation(T_c)$$

By sequentially applying crossover and mutation on selected elite parents, we can generate a new population of diverse CoT promptings while maintaining performance quality and ensuring stability in the evolution process.

3.3.2 Crossover Operation

To perform the crossover operation on parent1 and parent2 we separate each prompt into an "instruction" and "steps" component. With 50% probability, the operation combines instructions from parent1 with steps from parent2, otherwise visa versa to create the offstring CoT prompt.

3.3.3 Dual-Mutation System

We introduce a sophisticated mutation operation that uses the LLM to analyze and interpret the performance of the prompt, focusing on meaningful improvements rather than random mutations. The Intelligent-Mutation method begins with an analysis prompt that queries GPT-4o Mini for feedback on the given prompt T_c, the generated response, and performance metrics including semantic similarity and numeric accuracy. Specifically, the analysis prompt asks the LLM to identify a list of key issues, specific recommendations, and key areas that need emphasis (Full analysis prompt in Appendix).

We predefine common issue patterns along with their corresponding triggers and enhancements:

- Counting (triggers: imprecise counts, missed items, counting error): Enhancements focus on systematic counting methods, like dividing images into grids and counting sequentially.
- Verification (triggers: no verification, accuracy issue, inconsistent counts): Adds steps to double-check and compare counts for accuracy.
- Structural (triggers: confusing steps, poor organization, unclear instructions): Organizes prompts into clear phases (e.g., assessment, counting, verification).
- Uncertainty (triggers: unclear visibility, assumption made, guessing): Encourages explicitly addressing visibility issues, limitations, and providing confidence levels.

Based on the LLM's analysis, we then apply targeted mutations at a 30% rate by splitting the prompt into lines and adding corresponding enhancements based on identified issues. If missing, new sections like "Verification Steps" or "Confidence Reporting" may be appended if emphasized in the analysis. Existing lines that don't conflict with the enhancements are preserved.

To ensure robustness, if no specific issues or suggestions are identified, the method falls back to a deterministic mutation method, which randomly applies an alternative emphasis word with 30% mutation rate, and verification word with 20% mutation rate. If 'count' is present in the prompt, we replace it with a random alternative counting word (see Appendix for emphasis, counting, and verification word lists).

4 Results

We perform the evolutionary CoT algorithm with a fixed population size of 20 prompts per generation and three generations by default. The top 20% (4 best-scoring prompts) are preserved as elite prompts between generations and best performing prompts are tracked for each image in the JUS dataset. The highest-scoring result is recorded across multiple runs.

4.1 Evaluation Scoring

To calculate the fitness score of a prompt, we utilize two performance metric: semantic similarity weighted at 70% and numeric accuracy weighted at 30%. The semantic similarity score is calculated by generating embeddings with the sentence-transformers model to calculate cosine similarity between the model's response and the expected answer. In addition, a penalty is applied for responses that are too short or too long.

The numeric accuracy score applies only to counting questions and evaluates how accurately a generated rationale predicts the target numeric value. It handles "impossible to count" cases by checking for these phrases in the rationale and target text, and applies a tolerance for small numeric deviations. Errors are penalizes proportionally beyond the threshold to calculate the accuracy score.

4.2 Benchmarked Results

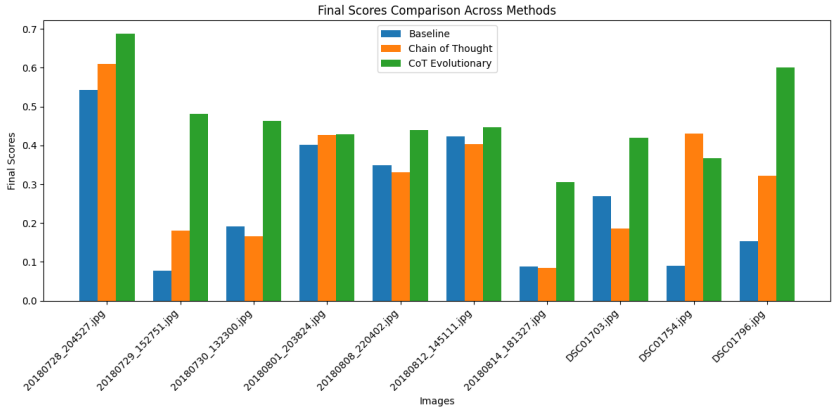

Figure 1: Evolutionary CoT prompting strategy displays superior performance across images when benchmarked against Zero-Shot CoT and Baseline Querying

We compare the final evaluation score for the prompt selected by our evolutionary CoT algorithm and compare it to the performance of our standard CoT prompt and the baseline direct query approach. Figure 1 shows that in all scenarios, advanced EoT performs significantly better than the baseline approach, whereas standard CoT actually performs worse that the baseline for half (5/10) the image recognition tasks. Comparing EoT to CoT scores, we examine that EoT still performs better in 9 out of 10 cases, with nearly half displaying over a 100% improvement in scoring.

However, the analysis also underscores the occasional limitations of even advanced prompting methods, as evidenced by specific outlier cases like the "DSC01754.jpg" image, where standard CoT outperforms EoT. This image displays 5 octopuses obscured by Japanese characters in the foreground, background, and details of a photo with the task being to count the number of octopuses. We discover that the EoT response resulted in an error because the LLM "could not analyze or view the image" and never ends up addressing the original question and outputting any numbers. These instances indicate that while EoT improves overall reliability, VLMs remain prone to unpredictable errors under certain conditions. Future work could focus on augmenting the evolutionary framework with additional robustness checks or ensemble methods to further mitigate such anomalies.

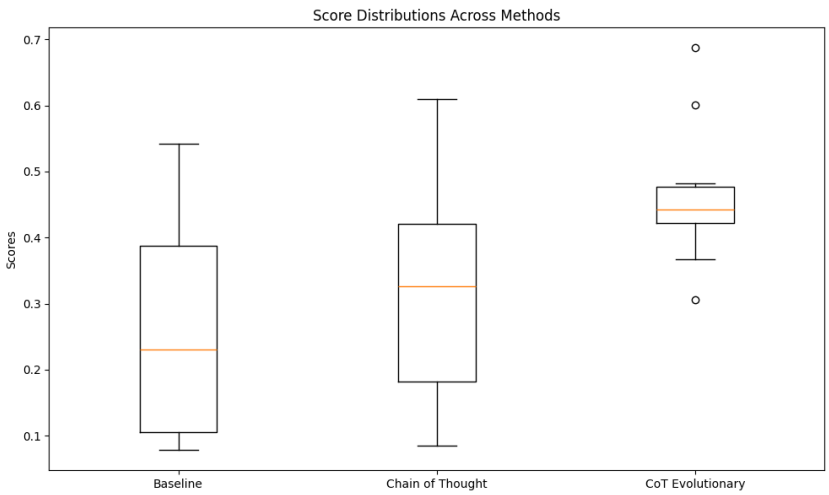

Figure 2: EoT Prompting results in drastically reduced variance and an overall higher mean score.

In Figure 2. the results demonstrate the robustness and effectiveness of the EoT prompting method in addressing reasoning challenges as EoT achieves both higher mean performance and significantly reduced variance compared to standard CoT prompting and baseline querying methods. This reduction not only improves prediction stability, but suggests that the guided mutation process effectively aligns prompts with task-specific requirements, leveraging labeled ground truth values to iteratively enhance performance.

While the upper boundary of the EoT method is lower, the average final score improves by over 0.2 points from baseline method and around 0.1 points from standard CoT. Furthermore, the lower bound of EoT scores displays a nearly 0.4 point improvement from both other methods. Interestingly, there are also several outliers in the EoT method: two that are near or above the maximum CoT score, and one just below the CoT mean.

5 Discussion and Conclusions

The results of this study highlight the significant potential of EoT prompting to improve the reasoning performance of VLMs, particularly in complex and ambiguous tasks. The EoT framework demonstrates consistent gains over both baseline and static CoT prompting approaches by dynamically tailoring prompts through evolutionary optimization. These improvements are particularly pronounced in tasks requiring nuanced reasoning, such as object counting and image recognition of geographical locations, where the CoT prompting often falls short.

The key implication of these findings is the possibility of enhancing VLM performance post-training through prompt optimization. Traditional improvements to VLMs typically require extensive computational resources for retraining or fine-tuning. In contrast, the EoT approach leverages labeled ground truth values to iteratively refine prompts, effectively improving model outputs without altering the underlying model architecture or parameters. This represents a significant shift in how performance enhancements can be approached, reducing the dependency on expensive retraining pipelines while still achieving substantial accuracy gains.

One of the most impactful aspects of this approach is the use of intelligent mutation, which incorporates ground truth values to guide the evolution of prompts. By analyzing

the discrepancies between model-generated responses and labeled answers, the framework identifies specific weaknesses in reasoning and introduces targeted improvements to prompts. This process ensures that subsequent generations of prompts are better aligned with the task requirements, driving incremental improvements in model outputs. The ability to adapt prompts based on ground truth also introduces a feedback loop that reinforces the alignment of VLM reasoning with real-world expectations.

The results suggest that prompt optimization could serve as a practical and cost-effective alternative to retraining for improving VLM performance in specific domains or use cases. For example, organizations with access to labeled datasets can use this approach to fine-tune model behavior for specialized applications, such as medical image analysis, autonomous driving, or cultural artifact recognition, without incurring the high costs of retraining multimodal models. Moreover, the reduced variance in performance observed with EoT prompting suggests improved robustness and reliability, which are critical for deploying VLMs in high-stakes environments.

5.0.1 Limitations

Our work identifies several limitations that warrant further exploration to fully realize the potential of Evolutionary Chain-of-Thought (EoT) prompting for Vision-Language Models (VLMs). First, the compute-intensive nature of evolutionary algorithms poses challenges, particularly in scenarios requiring rapid inference. The current approach involves generating and evaluating multiple prompts per generation - due to time constraints, we were only able to evaluate 10 of the 39 images. Additionally, resource constraints restricted our experiments to a fixed CoT population size and a single VLM model (GPT-4o Mini). Future work should explore varying population sizes, alternative VLM architectures, and being able to scale to larger, more diverse datasets to generalize the findings further.

5.1 Conclusion

This study demonstrates the potential of large language models (LLMs) as powerful tools for optimization, particularly in enhancing reasoning capabilities through Evolutionary Chain-of-Thought prompting. By embedding LLMs into the evolutionary process as both optimizers and generators, we achieved significant improvements in vision-language tasks without the need for extensive model fine-tuning. These findings highlight the practicality and efficiency of EoT prompting, which iteratively tailors prompts to task-specific requirements, leading to more robust and accurate outputs.

Our approach aligns with the growing body of work integrating LLMs with evolutionary algorithms, illustrating their versatility in generating, evaluating, and refining solutions across various domains. The success of LLM-guided EoT prompting underscores its potential for industry applications and the interpretability of evolved prompts provide a viable alternative to traditional fine-tuning. As LLMs continue to evolve, their integration with optimization frameworks like evolutionary algorithms promises to drive innovation in reasoning-driven AI, unlocking new possibilities across a wide range of applications.

References

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Appendix

Base Prompts

Counting

Analyze this image systematically:
1. Divide the image into clear sections
2. Count items in each section methodically
3. Double-check each section
4. If counting is impossible, explain why
5. State your confidence level and any uncertainties

Provide final count or detailed explanation if counting isn't feasible.

Precise counting process:
1. Assess image clarity and item visibility
2. Use grid method: count left-to-right, top-to-bottom
3. Note any partially visible or obscured items

4. Verify count by counting in reverse
5. If counting is not possible, explain specific reasons

State final count or explain why counting cannot be done accurately.

Descriptive

Analyze this image step by step:

1. Describe what you can clearly see
2. Note any ambiguous or unclear elements
3. Address the specific question asked
4. Provide confidence level in your answer

Be specific and avoid assumptions.

Systematic image analysis:

1. Focus on relevant elements for the question
2. Describe visible details methodically
3. Address any limitations in visibility
4.Answer the question with available evidence

State confidence level and reasoning.

Intelligent Mutation: Analysis Prompt

Analyze this prompt and its performance:
Original Prompt: [prompt]
Response Generated: [previous response]
Performance Metrics:
- Semantic Similarity: [X.X]
- Numeric Accuracy: [X.X]
- Found Numbers: [X.X]

Please identify specific issues with the prompt:

1. Is it failing to elicit precise counting?
2. Is it missing important verification steps?
3. Are the instructions clear enough?
4. Does it handle uncertainty well?

Provide your analysis in this exact JSON format:
"issues": ["issue1", "issue2"],
"suggested improvements": ["improvement1", "improvement2"],
"emphasis needed": ["area1", "area2"]
Ensure your response is valid JSON with these exact keys.

Deterministic Mutation

emphasis words = ['carefully', 'precisely', 'systematically', 'thoroughly', 'methodically']

```
counting words = ['count', 'tally', 'enumerate', 'quantify', 'calculate']
verification words = ['verify', 'check', 'confirm', 'validate', 'ensure']
```